



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

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Abstract

Global health equity (GHE) continues to be a major concern. A major contributing factor is the inconsistency of data systems resulting in health inequities such as access to high-quality healthcare, especially in resource-limited contexts like Sierra Leone. This is a research project that proposes a data engineering framework powered by AI to facilitate the aim of GHE via multi-source health data integration for evidence-informed practice. Findings will be reported in the healthcare context of Sierra Leone, characterized by high maternal mortality, lack of support in infrastructure and endemic health vulnerability post Ebola. The framework addresses fragmentation of data, the time it takes to detect outbreaks and the inequitable distribution of resources across rural and urban areas.

Specifically, the objectives of this framework are to: Design scalable data pipelines to allow for the real time secure data integration using tools such as Apache Airflow and Kafka; Develop AI Models using PyTorch and predictive analytics to detect inequities, forecast outbreaks and support resource allocation; and develop a visualisation platform with dashboards built on Django. The framework will prioritise ethics; for example, data privacy through General Data Protection Regulation (GDPR) compliance will be key along with community engagement to build trust, and scalability in data systems where (infrastructure) is low.

The significance of this study will highlight addressing gaps in technology and strategies for reducing health inequities that is supportive of the United Nations Sustainable Development Goals and forward looking for future application with Internet of Things (IoT) /nanotechnology. Expected outcomes of this research will entail operational pipeline, AI analytics layer, interactive platform for visualisation, and scalable architecture; ultimately providing a framework, articulating themes and creating a narrative for facilitating better health outcomes and equitable systems for Sierra Leone now, and for other countries in the years to come. This interdisciplinary approach leverage Python, MySQL/PostgreSQL, and scikit-learn to promote inclusive, data-driven global health advancements.

Introduction

Global health equity (GHE) is both the ethos and practice of making certain that every individual has a fair possibility to achieve their highest level of health, no matter their location, socioeconomic status, the colour of their skin, gender or any other distinctions. Global health is concerned with health issues crossing national borders which require international cooperation whereas health equity is about reducing unjust avoidable differences in health outcomes. Equity in healthcare means to provide care based on need and not based on the ability to pay or other biases and equity in global health is about assuring a fair distribution of healthcare resources.

GHE is essential in addressing the systemic inequities that create the conditions for millions of individuals to lack access to quality care and exacerbates issues of poverty and instability in low- and middle-income countries. GHE is especially urgent now given the state of global health today with pandemics like COVID-19, the impact of climate change, and the rising prevalence of non-communicable diseases all leading to widening gaps between wealthy and poor nations. GHE is particularly relevant in Sierra Leone, a low-income West African country that has been recovering from the Ebola epidemic that occurred between 2014-2016 and civil war, where GHE could lead to transformation of its already fragile healthcare system.

Sierra Leone's health care system is in crisis. By 2025, the health care system was under-resourced, and although the maternal mortality rate is down to 443/100,000 live births, it is still high. The health care system suffers from underdeveloped infrastructure, only a small number of modern hospitals are in urban areas (mostly in Freetown), while rural locations have no health care facilities: Health care in Sierra Leone also suffers from a lack of trained personnel (the doctor-to-patient ratio is approximately 1:10,000), a lack of effective supply chains for medicines, and the vulnerability to climate-related diseases like malaria and flooding. The 2021-2025 National Health Sector Strategic Plan attempted to implement universal health care coverage, but there are still issues such as overcrowding, sanitation, and socioeconomic factors that are preventing it from achieving that goal. Updating the SL health care system is in the implementation phase, and there are multiple WHO-supported programs - working on immunization, disease surveillance, and public health programs and national ambulance service launched in 2025 and funded through the SLMoH. Additionally, community health worker programs are also being implemented. Potential solutions include digital tools for integration of data, artificial intelligence for early outbreak prediction and insurance, international partnerships for funding, increasing educational and training opportunities, and implementing short-term research studies.

The progress of GHE through data engineering and AI can help unify fragmented data from various contributors such as hospitals, NGOs, and governments in order to assess the health status of populations in real-time. For instance, in Sierra Leone, it would allow for better resource distribution, early detection of outbreaks (cholera), and narrowing inequities. Globally, it would produce evidence-based interventions, working toward global initiatives like those outlined in the UN Sustainable Development Goals. Data engineering tools such as Apache Airflow used in ETL processes vehicle traditionally siloed data sources, and, when paired with unused or passive AI models (e.g. PyTorch machine learning) for predictive analytics, allow for the processing of multi-source, real-time data (electronic health records , as well as environmental health data through satellite data, and private data from social media accounts for sentiment analysis,). A mix of quality, relevant data sources would also be utilized; including but not limited to WHO datasets and facility-based national health surveys, as well as wearable IoT devices, patient self-reported tracking, and genomic datasets.

In low-resource settings like Sierra Leone, a significant challenge in adoption would come from real-time deployment caused by inconsistencies in electric power, inconsistent internet source, and the relatively low-level of skills of project implementers would need to be. But it can be scalable, requiring specific frameworks with practical data engineering pipelines for securely integrating across systems in an open-source manner, improved with AI capabilities that could then deploy disparity detecting algorithms, predict community-level outbreak risk, and guide prospective funding and resource distribution. In order to ensure this proposed solution is ethically deployable - promoting privacy (with GPDR specifications), building trust through community engagement, and measuring overall absence of net benefits for example from metric measurements of overall reduced mortality or morbidity rates from historically referenced population studies. Overall, future-proofed and ubiquitous integration of IoT/nanotech for continuous monitoring and surveillance, and to expand usability beyond Sierra Leone.

Problem Statement

Despite all advancements in technology, shared data in the Global Healthcare Ecosystem (GHE) is fragmented, detection of disease is delayed, and resources are unequally distributed across national health systems, all of which is a barrier to the development of GHE. Health records in Sierra Leone (SL) and elsewhere are stored within multiple silos, such as hospitals, research-based organizations, and governments, results in a lack of interoperability and informed decision-making. In SL the situation is acute; it has a health system that operates from a predominately paper based antiquated system, a limited digital infrastructure, and is data deprived due, in part, to a low literacy rate and limited connectivity.

Waning moments, such a practice unlike resourced settings, such as SL, face challenges in implementing data engineering and AI, including resource limitations (e.g. unreliability of power, limited computing hardware, etc), skills limitations (limited number of trained data engineers), and lack of available data: Africa does not have adequate digital infrastructure, old data centers, or fragmented datasets. The structural inequalities are limitations to AI implementation. Issues such a high cost of practice, unstructured data ecosystem, regulatory barriers. Data scientists and engineers rely on quality, structured data to train AI models: health data in SL is often missed or biased, and the data may also entrench existing inequalities.

Data pipelines must provide real-time processing, secure processing, and scalable integration. Current systems lack interoperability, therefore to design data pipelines requires integration of ETL tools to clean data across multiple sources (EHR, surveys), while considering bandwidth limitations and computational processing delays in SL. Security and ethics is important for sensitive data at the global and local levels, and SL, like all health ecosystems, is at risk for breaches, all while there are limited competent oversight frameworks. Community trust is low due to historical exploitation, requiring engagement strategies.

Research Objectives

The study aims to develop a data engineering and AI-driven framework for GHE, with specific objectives:

1. **Create and Deploy Data Pipelines:** Multiple sources of health data will need to be ingested, aggregated, and normalized through ETL with Apache Airflow, streamed in real-time via Kafka, and with interoperability through standards (e.g., FHIR). In low-resources settings, pipelines will accommodate intermittent connectivity at the edge with edge computing technology.
2. **Integrate AI/ML Models:** AI and ML will support the integration of models like those in scikit-learn for disparity detection (e.g., clustering underserved populations), PyTorch's deep learning for outbreak prediction (e.g., hybrid models integrating mobility data and web searches), and optimize allocation. Examples of integration can be found with tools like CAD4TB for detecting TB.
3. **Build a Visualization Dashboard:** A visualization dashboard could be developed with a web-based framework based in Django and include integrations with Plotly/Power BI for real-time indicators of equity to access data trends and disparities.
4. **Minimize Privacy, Security, and Ethics Risks:** Ensure clear measures are in place for appropriate GDPR/HIPAA regulations (e.g., transparent algorithms), incorporation of differential privacy and engagement strategies in communities for trust in the SL, and local engagement.
5. **Ensure Scalability and Flexibility of Framework:** Ensure the architecture accounts for scaling (Global) and providing flexible or adaptive infrastructure of a cloud-edge hybrid SL, and for IoT or Nanotech future integration with MQTT for real-time point-sensor data collection.
6. **Evaluate Impact and Future-ready the Framework:** Develop experiential metrics (PROMs, readmission rates) for outcomes, coupled with AI increasing the ability to mitigate bias, and ensure the framework iterative evolves technologies as necessary (e.g., responsible AI from the Global South).

Justification

The study is important because it employs data engineering, data science, and artificial intelligence, which show promise for breaking down data silos and deriving actionable insights for GHE. By analysing in the same space, it supports governmental, non-governmental, and non-profit agencies to identify discrepancies, model risks, and allocate resources to address needs, possibly reducing SL's maternal mortality even further while also progressing towards global equity.

Its significance stems from GHE being sustainable development's centrepiece and that without it, the poorest of the poor will disproportionately suffer in times of pandemics, as we witnessed in COVID-19. Furthermore, in SL, it addresses opportunely critical problems of climate-vulnerable health systems, while contributing to the 2022-2025 WHO Cooperation Strategy in recognizing the importance for all countries of using more real-time surveillance data to improve their health care systems.

By developing GHE's contribution with the proposed framework, it is expected that this will help globally via developing epidemic intelligence with AI-driven models (e.g., hybrid models enhancing forecasting), as well as locally, by merging all of the disjointed data into one space, which will be useful for targeting interventions. It can be argued that this approach future-proofs SL via a planned Internet of Things/nanotech points for real-time monitoring, as the technologies grow and networks mature.

The significance also contributes economic benefits; by applying efficiency in targeting allocations, resource saves will be achieved. In addition, it contributes to equity, with a reduction in bias on AI recommendations. Ultimately, it engenders research capacity and fills gaps for developing GHE in a low-resource setting, where we know skills deficits present barriers, to address capabilities to gather and use the data at scale.

Scope of the Study

The focus of this field is the creation of a framework for GHE, which emphasizes data engineering (real-time processing via Airflow/Kafka) and AI (models for differences/outbreaks) in SL, scalable to all settings. The data sources included in the scope are EHRs, surveys, IoT simulations; challenges in low-resource settings; privacy through encryption; going through the community; impact assessment (e.g. outcome metrics, equity indices).

Geographically, the focus is primarily on SL but could extend to similar settings (e.g. Africa). Temporally, the focus is on the current and into the future (e.g. integrating IoT). Excluding fully performed hardware deployment; utilizes simulated data.

Limitations: Resource constraints may impede testing; assumes moderate uptake.

Expected Outcomes

Outputs include: A functioning pipeline for integrating data; AI analytics for pattern/outbreak/equity; Web dashboard for visualization; A scalable architecture with IoT readiness; Contribution to equitable systems, measured one outcome at a time and decreasing inequities in SL and elsewhere.